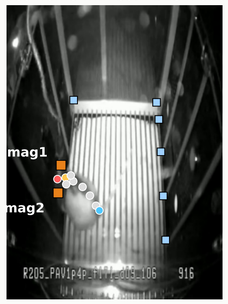
DLC magazine analysis: what each step does

Input: DeepLabCut tracking of the rat (9 body points) and of the box (calibration points), plus the trial table. Output, per video frame: at the magazine · head in the magazine · whole-body speed · head-anchored rotation; then per trial and per cue type. Example used in every box: Cohort01 R205, fixed ITI, Day 9 (99 trials, 78 min of video).

1 Inputs: video + two DLC files

Body DLC ('BehSyl' model): snout, neck, back1-4, tailbase, paws, per frame.
Calibration DLC ('NewMar5'): static box points, incl. mag1 / mag2 = the two ends of the magazine opening (median over confident frames).
Trial table: video frame of each cue onset (TrialLightOn_FrameIndex).

video frame with both DLC models



file	what
BehSyl.csv	rat, 9 points
NewMar5.csv	box points
trials.csv	cue-onset frames
video.mp4	optional (height, overlay videos)

2 Pixels -> floor cm -> magazine frame

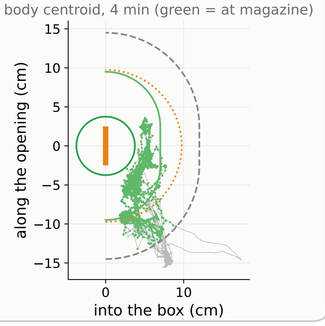
Camera views the box at an angle: 1 px = 0.18 cm wide but 0.11 cm tall, so pixels go through a 2x2 matrix (PX_TO_CM), not one factor.
Then everything is put in a frame centred on the magazine opening: x = cm into the box, y = cm along the opening.

cm = PX_TO_CM @ px

```
PX_TO_CM = [[0.1811, 0.0071],
            [-0.0144, 0.1080]]
```

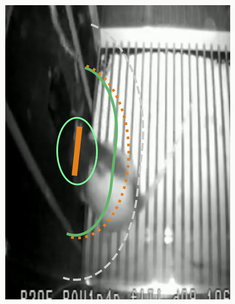
this session
opening length: 4.99 cm
box width check: 135.5 px
(reference 135.1 px ± 3%: ok)

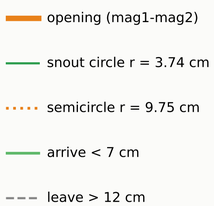
body centroid, 4 min (green = at magazine)



3 The zones around the magazine

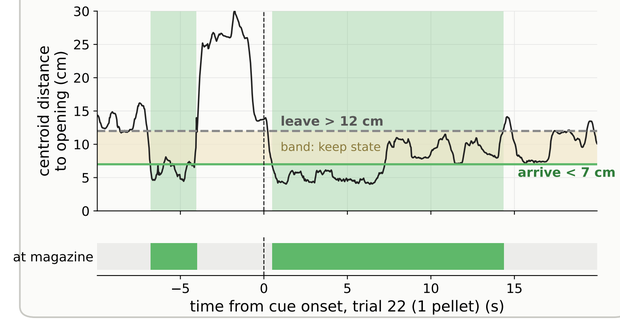
In cm the zones are circles; on the angled image they become ellipses.
Green circle (0.75 x opening) + orange semicircle (9.75 cm): head in magazine.
Solid green line (7 cm) + dashed grey line (12 cm): body at the magazine.
Distances are to the opening as a segment, not to its midpoint.





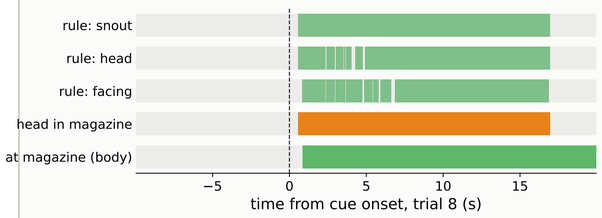
4 At the magazine (body)

Body centroid = mean of the 9 body points (low-confidence frames filled in).
The rat ARRIVES when the centroid is < 7 cm from the opening and LEAVES only when it is > 12 cm away; in between it keeps its state (hysteresis), so wobbling around one radius makes no fake visits. -> 'time at magazine'



5 Head in the magazine (entry rule)

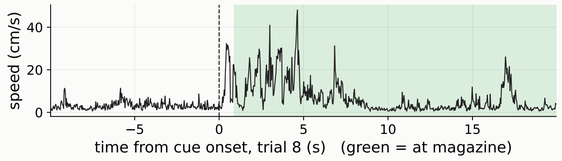
A frame is an entry when ANY of three rules holds (likelihood >= 0.8):
  snout: snout inside the green circle
  head: neck->snout segment touches it, and back1 in the semicircle
  facing: back1 in the semicircle, back1->neck line points at the opening

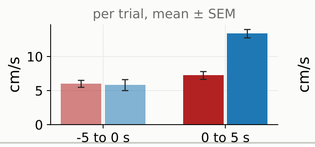


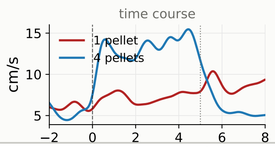
Whole session: head in magazine 19.3% of frames,
body at magazine 29.4%. The body definition is broader (the rat can stand there with its head out).
Rotation (boxes 7-8) uses the body definition.

6 Whole-body speed

Per body point: frame-to-frame step (through PX_TO_CM) x 29.97 fps.
Kept only if both frames have likelihood >= 0.8 and speed <= 150 cm/s (bigger jumps are DLC label swaps). Whole-body speed = mean of the usable points, if at least 3 are usable. Per trial: mean over -5-0 s and 0-5 s.

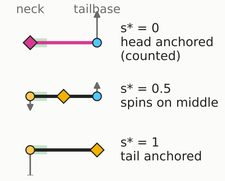


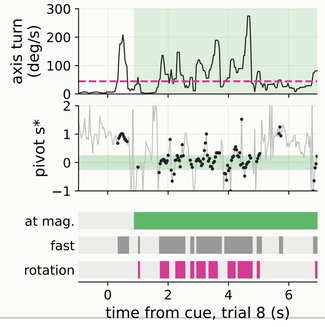




7 Head-anchored rotation (per frame)

Head stays at the magazine, body swings around it. A frame counts when ALL hold:
  1 at the magazine (box 4)   2 neck->tailbase axis turns > 44.6 deg/s
  3 pivot s* (axis point that moves least) in the neck quarter: |s*| <= 0.25
Cut calibrated on hand-annotated turning syllables (AUC 0.86). Anchor = neck.





Right, top: axis turn speed; pink dashed = the cut.
Middle: pivot s*; green band = neck quarter; black dots = fast frames.
Bottom: rotation = at magazine + fast + dot inside the band.

8 Rotation as bouts

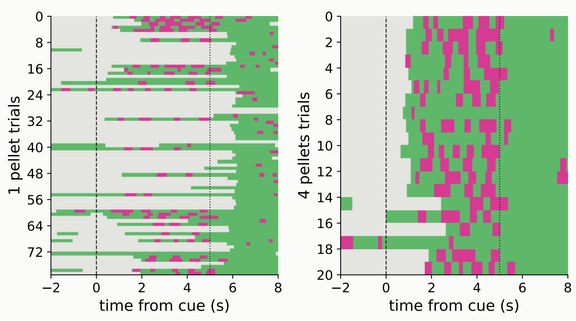
Single-frame labels flicker, so rotation is counted as bouts, made on the whole session before trials are cut: 1 gaps <= 5 frames (167 ms) are closed; 2 bouts < 5 frames (167 ms) are dropped; 3 bout frames must be at the magazine. Every rotation number is based on bouts.



Whole session: 1,013 per-frame runs -> 320 bouts.
Rotation share of magazine time: 8.8% per frame, 8.3% as bouts.
High / low rotation rats: fITI bout share (0-5 s, mean of 1p and 4p) above / below 15%. This rat: high.

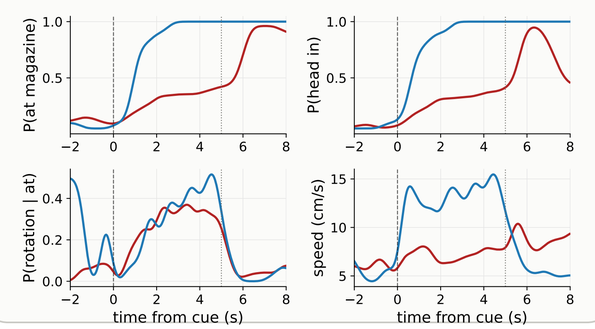
9 Cut trials around cue onset

Every frame label is cut from -2 to +8 s around each cue onset (the trial table's video frame). Trials whose window leaves the video are dropped.
One row per trial: grey = away, green = at magazine, pink = rotation bout.
Dashed line: cue onset. Dotted line: pellet (5 s).



10 Time courses (1 pellet vs 4 pellets)

Per frame offset, across trials: P(at magazine), P(head in magazine), P(rotation | at magazine) and mean speed. Numerator and denominator are Gaussian-smoothed separately (sigma 0.26 s), so frames with few trials at the magazine do not dominate. Red = 1 pellet, blue = 4 pellets.



11 Numbers per trial and per cue type

conditions.csv: 0-5 s after cue onset, frames pooled over trials (one row per cue type; these are the per-rat values used in the group analyses).
trials.csv: the same measures for every single trial, plus speed before and after the cue. frames.csv.gz: every label for every frame.

measure (0-5 s after cue)	1 pellet	4 pellets
trials	79	20
at magazine (% of 0-5 s)	29.5	78.5
head in magazine (% of 0-5 s)	28.6	83.4
rotation (% of magazine time)	30.2	35.9
rotation (% of 0-5 s)	8.9	28.1
rotation (deg per s at magazine)	39.7	44.1
speed (cm/s)	7.2	13.4

Check: with the full-precision DLC file these per-cue values are identical to the published table of the original analysis (e.g. 4 pellets: 78.5% at magazine, 35.9% rotation). The example file is rounded to 0.01 px, which moves 1p rotation from 30.0 to 30.2%.

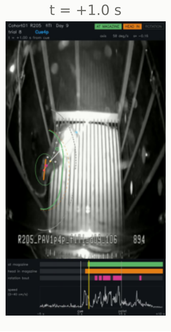
12 Overlay videos: check it by eye

make_trial_video.py draws the zones, the DLC skeleton, the centroid (green = at magazine) and the body axis (pink = rotation bout; diamond = pivot s*) on each trial, with state badges and a timeline. Below: trial 8 (4 pellets), before the cue, approaching, and during a head-anchored rotation.

t = -2.0 s



t = +1.0 s



t = +3.2 s



```
Run it:   python analyze_session.py --body-dlc <rat DLC> --calibration-dlc <box DLC> --trials <trials.csv> --out <folder>      python make_trial_video.py ... --video <mp4> --trial 8 --out <folder>
```

Same code and numbers as the Dopamine-and-Timing analyses: on this session every per-frame label (distance, at magazine, head-entry rules, speed, axis turn, pivot, rotation frames and bouts) matches the original modules on all 140,055 frames. All numbers live in dlc_magazine/settings.py.